Tarun Sai Madimi

Full Stack Developer

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Full Stack Developer with 3+ years of experience building scalable web applications, APIs, and event-driven systems using Java, Python, TypeScript, and cloud-native technologies. Delivered platforms supporting 99.9% service availability and processing over 10M+ transaction and event records annually across fintech and cybersecurity environments. Experienced in microservices, distributed systems, real-time data pipelines, cloud infrastructure, and secure application development using AWS, Azure, PostgreSQL, Kafka, and modern frontend frameworks.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, SQL
- **Backend & API Development:** FastAPI, Celery, Spring Boot, Spring WebFlux, Spring Batch, Express.js, Node.js, Hibernate ORM, SQLAlchemy, R2DBC, Apache Camel, GraphQL, Strawberry, RESTful APIs, gRPC
- **Frontend Development:** React, Next.js, Redux Toolkit, React Query, WebSockets, Server-Side Rendering (SSR)
- **Data Engineering & Machine Learning:** Apache Kafka, Apache Spark, scikit-learn, ETL Pipelines, Real-Time Streaming
- **Databases & Cloud Technologies:** PostgreSQL, MongoDB, Redis, AWS (Lambda, API Gateway, SQS, S3, CloudWatch) Azure AKS
- **DevOps, Monitoring & Quality Engineering:** Terraform, Docker, Helm, Jenkins, GitHub Actions, OpenTelemetry, Grafana Tempo, ELK Stack, SLF4J, React Testing Library, JUnit, Mockito, SonarQube, ESLint, Blue-Green Deployments, Infrastructure as Code (IaC)
- **Security, Cybersecurity & Architecture:** AWS Cognito, JWT Authentication, OAuth2, RBAC, OWASP Security Practices, VirusTotal API, MISP API, SIEM, Threat Intelligence Enrichment, Alert Correlation, Adversarial Simulation, Microservices, Multi-Tenant Systems

PROFESSIONAL EXPERIENCE

Full Stack Developer | Amazon **Oct 2024 – Present**

- Contributed to RESTful microservices using FastAPI and Celery with async task queues, integrating internal APIs to automate data enrichment workflows and cut manual processing time by 50%.
- Took ownership of a real-time event correlation engine using Python asyncio and Amazon SQS, aggregating multi-source data streams into unified event timelines and reducing duplicate alert noise by 60% for internal operations teams.
- Collaborated with backend teams to expose normalized operational data through a GraphQL API built with Strawberry and SQLAlchemy, deployed behind AWS API Gateway with rate limiting and authentication to support multiple internal consumers.
- Drove automation of compliance and reporting workflows using Python scripting integrated with Terraform and AWS Lambda, cutting report turnaround time from 2 days to 4 hours.
- Worked alongside the observability team to instrument distributed backend services using OpenTelemetry and AWS CloudWatch, contributing to trace-level analysis that resolved key latency bottlenecks in production payment workflows.
- Partnered with product and design to rebuild internal operational dashboards in Next.js and TypeScript with live WebSocket data feeds, reducing UI regressions by 35% through improved testing and CI practices.

Software Developer | FIS Global **Nov 2021 – Aug 2023**

- Built a multi-tenant payment reporting framework using Java 17, Hibernate ORM, and PostgreSQL that consolidated client reporting pipelines for banking clients, reducing data processing time by 40% across enterprise accounts.
- Developed a real-time audit trail system using Spring Batch to manage ETL workflows between core banking systems and cloud data warehouses, supporting PCI-DSS compliance for financial institution clients.
- Designed RESTful microservices using Spring WebFlux and R2DBC to handle non-blocking, high-frequency transaction requests, improving service throughput by 45% during peak settlement and reconciliation windows.
- Refactored inter-service communication between payment gateway and reconciliation modules from synchronous REST to event-driven messaging using Kafka, reducing processing latency and improving fault tolerance across transaction workflows.
- Managed cloud infrastructure on Azure AKS using Terraform and Helm charts, enabling automated environment provisioning and zero-downtime deployments for production banking applications.
- Set up CI/CD pipelines using Jenkins and GitHub Actions, integrating automated build, test, and deployment stages that streamlined release cycles and improved team delivery consistency across environments.
- Enforced SonarQube code quality gates across all service repositories, achieving 85% test coverage through JUnit and Mockito which reduced production defects by 30% across the team.

PROJECTS

Insurance Claims Management Portal

Tech Stack: Spring Boot, React, TypeScript, React Query, PostgreSQL, JWT Authentication, Redis, Docker, AWS S3

- Architected a claims management portal with RBAC, automated validation workflows, OWASP-compliant API security practices, and caching strategies, reducing claim processing time by 45% and improving operational throughput.

Private Self-Destructing URL Shortener

Tech Stack: React, TypeScript, Express.js, MongoDB, OAuth2, AWS Cognito

- Modernized a privacy-first URL shortening platform with time-bound automatic link expiration and secure user authentication, enabling organizations to share sensitive resources with controlled access windows and zero manual cleanup overhead.

E-Commerce Web App

Tech Stack: React, Redux Toolkit, Node.js, MongoDB, Stripe, Firebase

- Delivered a high-performance e-commerce platform featuring product discovery, cart functionality, and secure payments, accelerating page loads by 40% and ensuring a seamless multi-device experience.

EDUCATION

Master of Science in Information Systems, Indiana Institute of Technology, USA **Jan 2024 – May 2025**